



LLM-based Academic Writing with Fine-Grained Citations

Master Thesis Presentation

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ScholarCopilot Overview

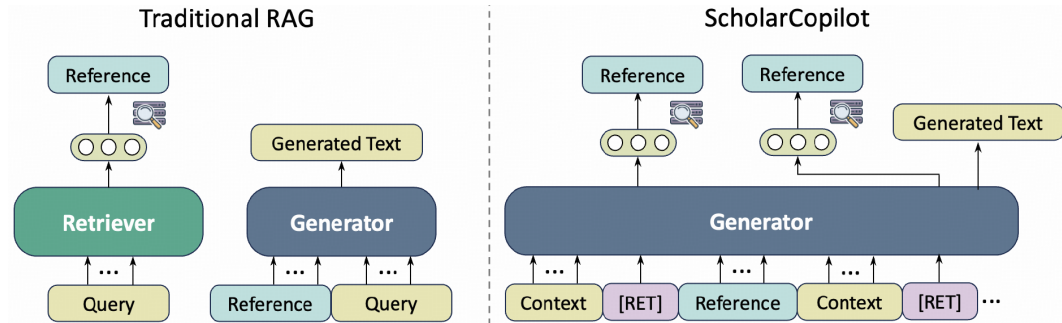


Figure: From [4] (modified).



Research Question

Does retrieving most relevant **passages** instead of abstracts improve ScholarCopilot regarding citation recall or generation quality ?

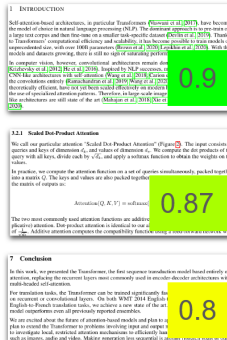
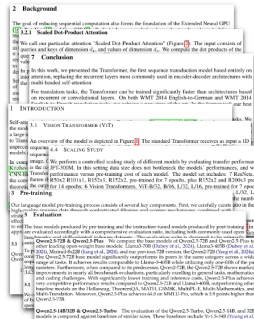


Ranking by ScholarCopilot
based on abstracts

Passages from documents
retrieved by ScholarCopilot



Reference Retrieval with the Passage Retrieval Module (continued)



Passages from documents
retrieved by ScholarCopilot

Ranking by the PR module
based on passages

Reference retrieved by the
PR module



Models

- ScholarCopilot: ScholarCopilot-v1 [4]
 - backbone: Qwen2.5-7B [5]
 - fine-tuned on the 500k ScholarCopilot training dataset
- Passage Retriever: Qwen2.5-7B-Instruct [5]
- Generation Evaluation model ("judge"): DeepSeek-R1-Distill-Qwen-7B [1]

Task 1: Reference Retrieval

Given a citing context, e.g. [2]:

Transformers were proposed by Vaswani et al. (2017) for machine translation, and have since become the state of the art method in many NLP tasks. Large Transformer-based models are often pre-trained on large corpora and then fine-tuned for the task at hand: BERT (<|TARGET-CITATION|>

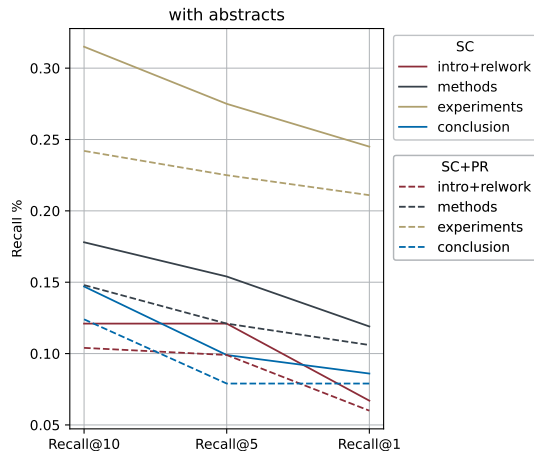
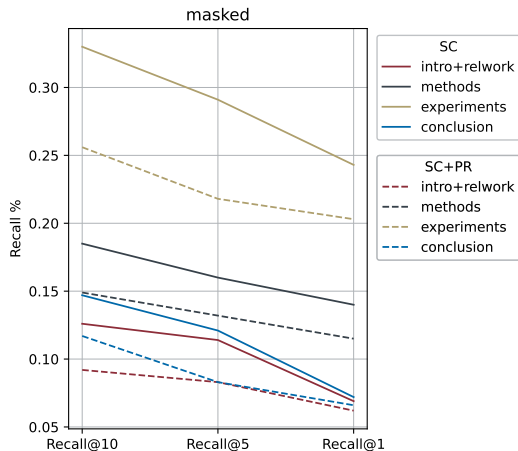
the model is tasked to retrieve the cited document.

Other citations – here "Vaswani et al. (2017)" – are either

- masked or
- replaced with the abstract of the cited document



Citation Recall Performance





Recall@ k decline towards small k

Model	Dataset	intro+relwork	methods	experiments	conclusion	Σ
SC	masked	-0.06	-0.04	-0.09	-0.07	-0.26
SC	with-abstracts	-0.05	-0.06	-0.07	-0.06	-0.24
SC+PR	masked	-0.03	-0.03	-0.05	-0.05	-0.16
SC+PR	with-abstracts	-0.04	-0.04	-0.03	-0.04	-0.15

Table: Cumulative Recall@ k difference from $k = 10$ over $k = 5$ to $k = 1$.



Why is the recall for experiments so much higher?

Maybe citations in experiments are easier for the model to resolve?

Consider this example from [2]:

We transfer the models trained on these dataset to several benchmark tasks: ImageNet on the original validation labels and the cleaned-up ReaL labels (Beyer et al., 2020), CIFAR-10/100 (Krizhevsky, 2009), Oxford-IIIT Pets (Parkhi et al., 2012), and Oxford Flowers-102 (Nilsback & Zisserman, 2008).

	intro+relwork	methods	experiments	conclusion
last token in abstract	0.08	0.12	0.22	0.11

Table: Fraction of cases where the last token before the citation appears in the abstract of the cited document.

Correct Retrieval Overlap

Dataset	Metric	intro+relwork	methods	experiments	conclusion	μ
masked	Recall@10	0.91	0.97	0.98	0.91	0.943
masked	Recall@5	0.94	0.94	0.96	0.94	0.945
masked	Recall@1	0.79	0.89	0.88	0.88	0.86
with-abstracts	Recall@10	0.9	0.93	0.96	0.95	0.935
with-abstracts	Recall@5	0.86	0.97	0.95	0.89	0.918
with-abstracts	Recall@1	0.87	0.87	0.93	0.84	0.878

Table: Fraction of correct retrievals by ScholarCopilot given correct retrieval by the PR module.



Task 2: Generation Quality Evaluation

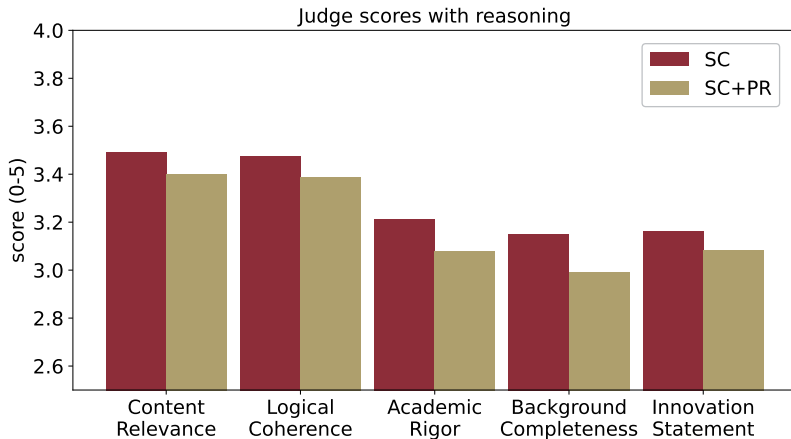
Given

- title
- abstract
- the first 2,5 sentences of the introduction

of an existing paper, the model is tasked to generate a scientific paper.

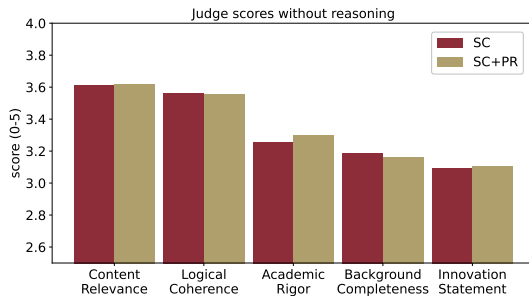
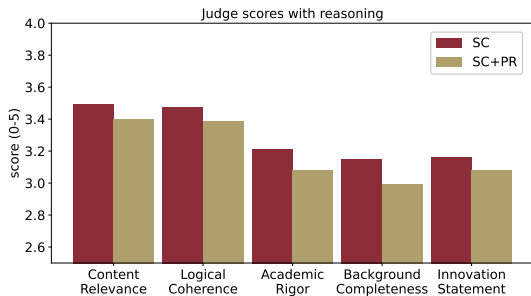


Generation Quality Evaluation: Results



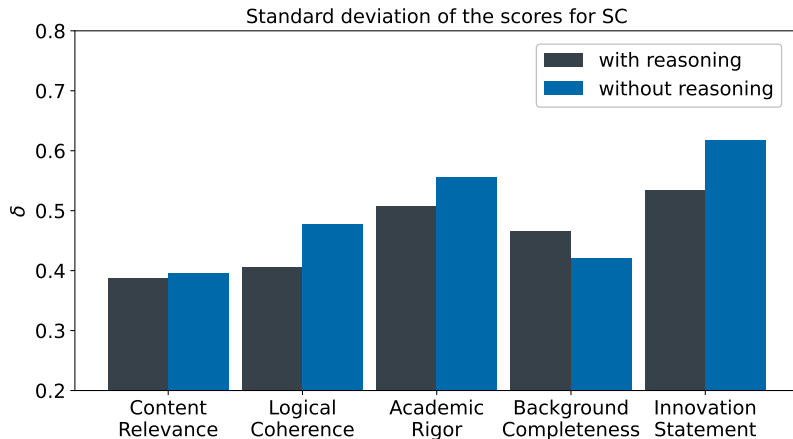


Generation Evaluation Comparison: With vs. Without Reasoning





Reasoning Coincides With Lower Standard Deviation



Conclusion

- the PR module does **not improve ScholarCopilot** in either citation recall or generation quality
- the PR module is able to **identify correct citation targets**, even if ScholarCopilot fails to do so
- citations in experiments sections tend to be **easier to retrieve** compared to other sections
- LLM-based generation evaluation shows more consistent scoring behaviour when prompted to **provide reasoning** for its decision



Passage Retrieval with ReasonIR

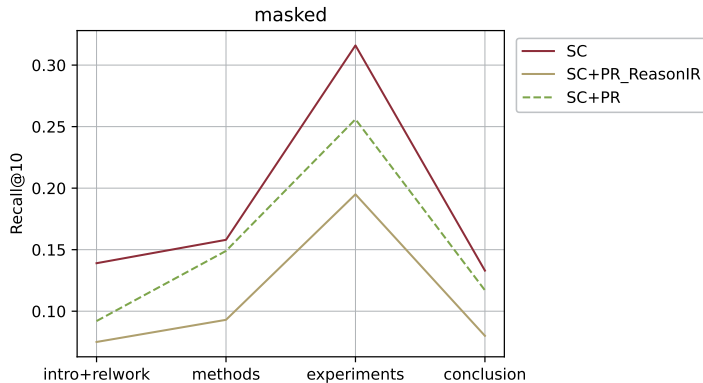


Figure: Citation recall when retrieving passages with ReasonIR [3] and reranking with Qwen.

- [1] DeepSeek-AI et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. In *arXiv preprint arXiv:2501.12948*.
- [2] Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit & Neil Houlsby. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. In *arXiv preprint arXiv:2010.11929*.
- [3] Shao, Rulin, Rui Qiao, Varsha Kishore, Niklas Muennighoff, Xi Victoria Lin, Daniela Rus, Bryan Kian Hsiang Low, Sewon Min, Wen-tau Yih, Pang Wei Koh & Luke Zettlemoyer. 2025. ReasonIR: Training Retrievers for Reasoning Tasks. In *arXiv preprint arXiv:2504.20595*.
- [4] Wang, Yubo, Xueguang Ma, Ping Nie, Huaye Zeng, Zhiheng Lyu, Yuxuan Zhang, Benjamin Schneider, Yi Lu, Xiang Yue & Wenhua Chen. 2025. Scholarcopilot: Training large language models for academic writing with accurate citations. In *arXiv preprint arXiv:2504.00824*.

- [5] Yang, An, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang & Zihan Qiu. 2024. Qwen2 technical report. In *arXiv preprint arXiv:2407.10671*.